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Glossary Term

Spatial Analysis

What Is Spatial Analysis?

Spatial analysis is a practical data evaluation method that explores how physical locations and geographic relationships directly impact everyday business results. This approach treats map coordinates and physical boundaries as measurable variables, uncovering the location-driven trends that standard reports usually miss.

Expanded Definition

Spatial analysis treats physical location as more than simply a dot on a digital map. It brings physical location into the analytical workflow as data that can be calculated, compared, and combined with other business information. The USGS data-management page describes spatial analysis as examining locations, attributes, and relationships in spatial data to answer questions and create new information.

It uses coordinates, distances, and areas — techniques traditionally rooted in geographic information systems (GIS) — to understand how physical location affects business performance. Analysts can visualize distribution networks, calculate drive times, and identify geographic concentrations of similar activity that might otherwise remain hidden in spreadsheets.

Unlike data visualization, which helps present geographic results, spatial analysis focuses on calculating relationships such as proximity and density. It's closely related to spatial analytics, but refers more specifically to the methods used within broader business intelligence workflows.

According to 2026 Fortune Business Insights research, spatial analysis represents a growing solution segment as businesses use these capabilities to understand customer behavior and demand. Grand View Research values the global geospatial analytics market at $117 billion in 2026 and expects it to reach $234 billion by 2033 as organizations adopt location-based services across retail, transportation, and logistics.

How Spatial Analysis Is Applied in Business & Data

Organizations apply spatial analysis when geographic proximity can add explanatory value to an existing business data set. Business analysts integrate location attributes into their performance reporting to evaluate things like regional market penetration or resource distribution. Data scientists frequently rely on these same calculated spatial variables as features in their predictive analytics models.

Analysts rely on these geographic capabilities for several core tasks:

- Data enrichment: Analysts associate customer records with specific geographic territories based on coordinate locations instead of relying on inconsistent text fields. The resulting spatial attribute standardizes regional reporting across the entire organization.
- **Predictive modeling:** Data scientists calculate proximity measures to use as explanatory features within their forecasting algorithms. This geographic variable helps determine whether physical distance strongly correlates with the modeled business outcome.
- **Resource planning:** Planners evaluate whether physical assets are positioned effectively in relation to the operational demand they support. They adjust their allocation strategies based on these spatial distribution patterns to improve service delivery.
- **Territory analysis:** Operations managers measure how customer accounts are geographically distributed across existing service regions. They use these density patterns to redraw physical boundaries and balance team workloads.

The Alteryx platform provides specific capabilities for this work by enabling teams to build geographic processing directly into their automated workflows, alongside tabular data.

How Spatial Analysis Works

Spatial analysis begins by connecting business data to a physical location. Spatial data can come from internal business systems or external geographic data sources. That location information may already exist as latitude and longitude coordinates, or it may be embedded in CRM addresses, store locations, GPS feeds, operational systems, or asset data. Analysts can also bring in external geographic reference data when they need boundaries, road networks, demographics, or other location context.

When source records contain addresses instead of coordinates, analysts may first use geocoding to connect those records to a physical location.

After the location data is available in a usable format, it can move through the spatial analysis workflow, which generally follows five steps:

1. **Prepare the location data:** Analysts validate addresses or coordinates and convert them into a consistent spatial format. This step may also involve combining internal business data with external geographic data.
2. **Represent the geographic features:** The workflow creates spatial objects that reflect what is being analyzed. A branch might be represented as a point, for example, while a sales territory could be represented as a polygon.
3. **Choose the spatial relationship:** Analysts define what they want to measure. Depending on the question, they might execute spatial joins to compare areas of geographic overlap, determine whether a location falls inside a territory (containment), identify the nearest facility (nearest-location analysis), generate buffers to build trade areas, or calculate proximity using straight-line versus road-network distance.
4. **Calculate new spatial attributes:** Spatial algorithms turn those relationships into usable data. A workflow might append a distance value, territory assignment, nearest-location identifier, or other geographic result to each record.
5. **Combine spatial and business analysis:** Analysts evaluate the new spatial attributes alongside measures such as revenue, demand, service levels, or risk. The spatial result can then feed reporting, segmentation, predictive modeling, or other downstream analytics.

Other common spatial analysis techniques include clustering and hotspot analysis to identify areas where similar activity is concentrated.

Common spatial analysis challenges

Analysts commonly encounter a few technical hurdles when working with geographic data sets. Addressing these issues early helps ensure that spatial metrics accurately represent the real-world relationships behind the analysis:

- **Data quality and geocoding errors:** An incomplete address or misspelled street name can produce an inaccurate geocode, placing a customer or asset in the wrong city or ZIP code.
- **Mismatched geographic representations:** Overlaying data from different sources can cause alignment issues if the files use different coordinate reference systems or outdated boundaries.

- **Routing vs. straight-line distance:** Calculating a simple straight line (as the crow flies) may tell a completely different story than measuring actual travel distance across a physical road network.
- **Computational limits:** Processing millions of complex spatial relationships — like generating overlapping drive-time polygons for a national retail network — is computationally intensive and requires robust processing power.

Addressing these challenges early helps analysts produce spatial results that more accurately reflect real-world geographic relationships.

Examples and Use Cases

Geographic calculations improve existing analytical workflows by providing physical context. Different business teams and functions use the resulting measurements to segment records and explain performance variations that aren't obvious otherwise.

- **Sales and marketing:** Campaign managers create spatial catchments — like drive-time boundaries — around retail locations to determine where their most valuable customers live and target regional promotions accordingly.
- **Supply chain and logistics:** Network planners calculate the travel distance or estimated drive time between distribution centers and final delivery points to balance inventory loads, reduce transportation costs, and optimize routing.
- **Real estate and site selection:** Expansion teams evaluate demographic density and competitor proximity surrounding proposed properties to forecast potential foot traffic and evaluate promising new store locations.
- **Risk management:** Insurance and compliance teams map physical corporate assets or policyholder locations against known environmental hazard zones to assess geographic exposure and adjust risk pricing.

Industry Use Cases

Different sectors adapt location intelligence to solve their specific operational and network challenges. Organizations rely on these geographic calculations to optimize physical asset placement and manage localized risks.

- **Financial services:** Fraud investigators evaluate the physical distance between consecutive transaction locations to identify impossible travel scenarios. This geographic calculation serves as a signal for flagging suspicious transactions for further review.
- **Healthcare:** Administrators measure the travel distance between residential populations and specialized urgent care facilities. They use this access data alongside demographic trends to plan future clinic expansions.
- **Manufacturing:** Supply chain directors map out vendor facilities to identify hidden dependencies within specific industrial parks or regions. This network visibility allows them to proactively source secondary suppliers — outside of concentrated geographic areas.
- **Telecommunications:** Infrastructure planners compare the physical placement of existing transmission equipment against areas with high bandwidth demand. Being able to see coverage gaps helps them estimate where new hardware investments may have the greatest impact.

Frequently Asked Questions

What is the difference between spatial analysis and GIS? While a geographic information system (GIS) refers to the specialized software and framework used to store, manage, and visualize geographic data, spatial analysis is the actual mathematical and analytical method used to uncover relationships within that data. Today, analysts can perform spatial analysis directly inside broader data platforms without needing a standalone GIS environment.

What is the difference between spatial analysis and spatial analytics? Spatial analysis refers to the specific mathematical methods used to examine geographic relationships within a data set. Spatial analytics represents the broader organizational practice of incorporating those calculated methods and their results into daily business intelligence operations.

What types of data can be used in spatial analysis? Any business record that can be associated with a physical location supports this type of evaluation. The location component might start as a coordinate pair, a standardized address, or an existing spatial object like a ZIP code boundary.

Does spatial analysis always require a map? No, analysts don't always need to render a visual map to perform complex geographic calculations. The process often produces numerical outputs, such as distance measurements or intersection flags, that feed directly into downstream statistical models or automated reports.

How is spatial analysis different from traditional data analysis? Traditional data analysis examines the relationships among numerical or categorical variables without considering where the observations actually occurred. Spatial techniques introduce physical location and proximity as core analytical variables that interact with standard business metrics.

Further Resources on Spatial Analysis

- **Webinar** | Supply Chain Intelligence: A Geospatial Approach to Optimal Decision-Making
- **White Paper** | Unify Your Geospatial Analytics
- **Blog** | An AI Playbook for Location Analytics in Retail
- **Documentation** | Alteryx Designer Spatial Tools

Sources and References

- **Grand View Research** | Geospatial Analytics Market Report
- **Fortune Business Insights** | Geospatial Analytics Market Research Report
- **National Geographic** | Geographic Information System (GIS)
- **Oracle** | Geocoding
- **U.S. Geological Survey** | Data Management: Analyze

Synonyms

- Geospatial analysis
- Geographic data analysis
- Location-based analysis

Related Terms

- Spatial Analytics
- Business Intelligence
- Data Visualization
- Data Analytics

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Alteryx Editorial Standards and Review This glossary entry was created and reviewed by the Alteryx content team for clarity, accuracy, and alignment with our expertise in data analytics automation.

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